



江苏省苏州中学国际部
Suzhou High School – International Division

SZZX-ID Bridge Program Physics

Newton' s law Problem Solving Practice Lesson

Class Outline

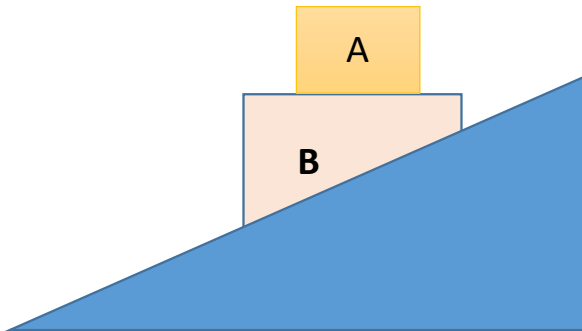
Title of Unit	Forces and Newton's Laws
Title of Lesson	Problem solving practice lesson
Starter Activity	Potato demonstration – the climbing potato
Direct Instructions	Teaching Strategies: Demonstration, Lecture, Collaboration, Questioning, Inquiry
	- Lecture: steps to solve a sample question - Inquiry: Kahoot challenge - Collaboration: work in groups for exercise
	Differentiation: Content, Instructional Strategies, the Classroom, Products, Teacher
	Teacher to give support to students during group work
	International Mindedness:
Lesson Objectives	By the end of the lesson, students will be able to: 1. Apply knowledge in force and Newton's laws to solve problems 2. Solve simultaneous equations 3. Work and study in a group
ATLs	Identify unit ATLs and how they are being addressed in the lesson
	Social: taking various roles in teamwork
IB Learner Profile	Link the lesson to the characteristics of the IB Learner attribute
	Communicator, thinker
Teaching Activities	Demo: climbing potato Kahoot challenge; Group exercise of problem-solving;
Plenary Session	Checking for understanding / Homework / Q & A / Reflection etc.
	Homework instruction: (known conditions), formula, substitution, calculation

Warm up – in the language you are familiar with



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A 物体和B物体一起沿着斜坡匀速下滑，则A物体受到的力是：



- A ✓ Force of gravity, Normal force from B to A
- B. Force of gravity, Normal force from B to A, Sliding force along the slope
- C. Force of gravity, Normal force from B to A, Force of friction
- D. Force of gravity, Force of friction, sliding force along the slope

Warm up – in the language you are familiar with



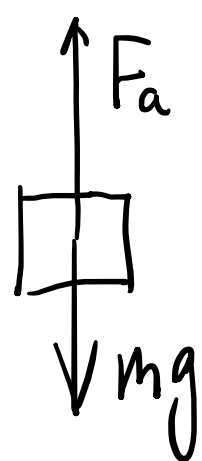
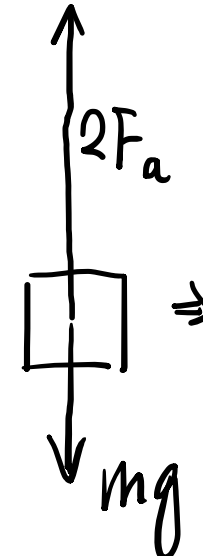
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竖直起飞的火箭在推力 F 的作用下产生 10 m s^{-2} 的加速度。当推力增大到 $2F$ 的时候，火箭的加速度会达到 (g 取 10 m s^{-2} ,忽略空气阻力)

- A. 20 m s^{-2}
- B. 25 m s^{-2}
- ☒ C. 30 m s^{-2}
- D. 40 m s^{-2}

Make sure to use free-body diagram to help you!

①


$$F_a - mg = ma_1$$
$$\Rightarrow F_a = m(a_1 + g)$$
$$= 20m$$

$$2F_a - mg = ma_2$$
$$\Rightarrow 40m - 10m = ma_2$$
$$\Rightarrow a_2 = 30\text{ m s}^{-2}$$

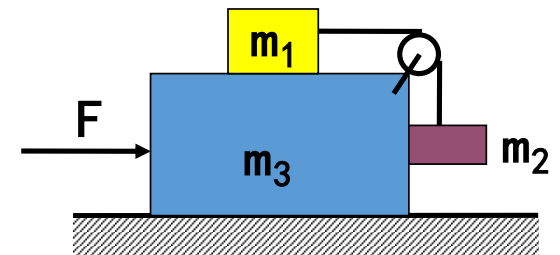
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如图三个物体的质量分别为 m_1 、 m_2 、 m_3 ，系统置于光滑水平面上，系统内一切摩擦和绳的重力均不计，求用多大的水平力 F 推才能使三物体间无相对运动。

Acceleration a is the same

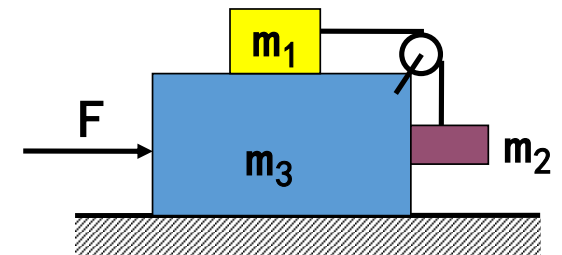


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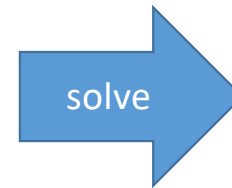
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Acceleration the 3 objects are the same, by Newton's second Law

$$F = (m_1 + m_2 + m_3)a$$

For m_1 , by Newton's second Law : $T = m_1 a$



$$F = (m_1 + m_2 + m_3) \frac{m_2}{m_1} g$$

For m_2 , equilibrium on the vertical direction gives: $T = m_2 g$

Activity – collaborate in a small group



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1. You will work as a group on a worksheet. Take turns for the following roles:
 - role 1: Read the question slowly to your group mates. Elaborate on the knowns and unknowns
 - role 2: draw FBD, and explain what laws/rule would be applicable
 - role 3: put down equations based on the known conditions and FBD
2. Once you put down the equations and ensure the equations are solvable, move to the next problem. You will finish the maths part at home
3. Note down the questions that none of you have clue of. Ask the teacher later.

ATL- Social

- Working collaboratively to achieve a common goal
- Assigning and accepting specific roles during group activities